Problem-Solution Fit

Date	29 June 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S)[CS] • Government Agencies • Meteorological Departments • District Administration • Emergency Response Teams • People living in flood-prone regions • Disaster Management Authorities	**6. CUSTOMER LIMITATIONS** EG.BUDGET,DEVICES **[CL]** • Limited access to real-time prediction systems • Budget constraints • Lack of advanced forecasting tools • Internet connectivity issues in rural areas • Limited technical expertise	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Available Solutions** • Traditional flood forecasting • Manual weather analysis • Basic rainfall monitoring systems **Pros** • Easy to implement • Widely used **Cons** • Lower prediction accuracy • Delayed warnings • Cannot efficiently analyze multiple weather parameters

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Delayed flood warnings
- Inaccurate flood predictions
- Heavy loss of lives and property during monsoon seasons
- Difficulty in planning evacuations
- Frequent occurrence during heavy rainfall periods

9.PROBLEMROOT/ CAUSE [RC]

- Extreme rainfall
- Climate change
- Insufficient early warning systems
- Delayed data analysis
- Dependence on conventional forecasting methods

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Monitor weather forecasts regularly during monsoon
- Depend on government alerts
- Evacuate only after official warnings
- High urgency during heavy rainfall

3.TRIGGERSTOACT [TR]

- Heavy rainfall
- High seasonal rainfall
- Poor visibility due to cloud cover
- Continuous weather changes
- Rising flood risk alerts

4. EMOTIONS
BEFORE / AFTER **[EM]**

Before

- Fear
- Anxiety
- Uncertainty
- Panic

After

- Confidence
- Preparedness
- Better safety
- Quick decision-making

10.YOURSOLUTION [SL]

Develop a **Machine Learning-based Flood Prediction System** using **Decision Tree, Random Forest, KNN, and XGBoost** algorithms. The system analyzes weather parameters such as annual rainfall, seasonal rainfall, cloud visibility, and other meteorological data to predict flood occurrence. The **XGBoost model achieved 96.55% accuracy** and is integrated into a **Flask web application** for real-time flood risk prediction. The application can be deployed on **IBM Cloud**, enabling disaster management authorities to receive timely predictions, issue early warnings, plan evacuations, and allocate emergency resources efficiently.

8.CHANNELSOF BEHAVIOR [CH]

ONLINE

- Web Application
- IBM Cloud Platform
- Government Disaster Management Portals
- Mobile and Weather Websites

OFFLINE

- Local Government Offices
- Disaster Response Teams
- Public Announcement Systems
- Community Awareness Programs